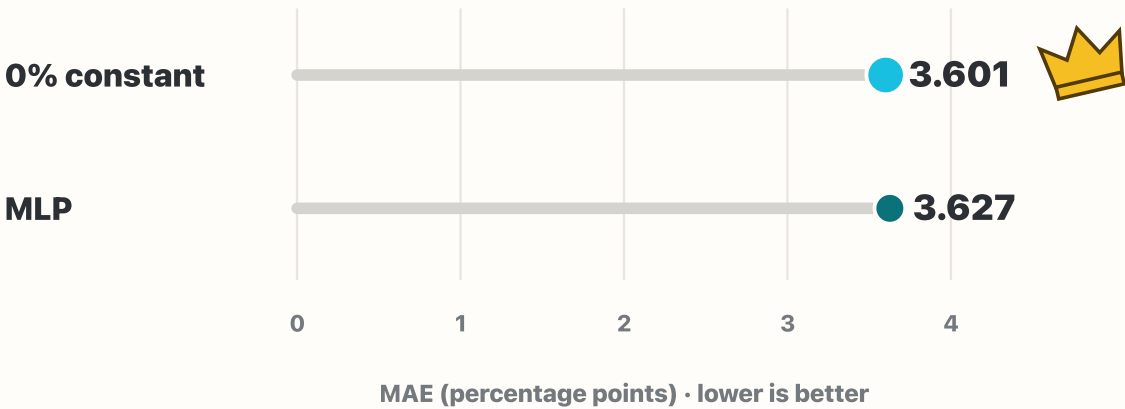
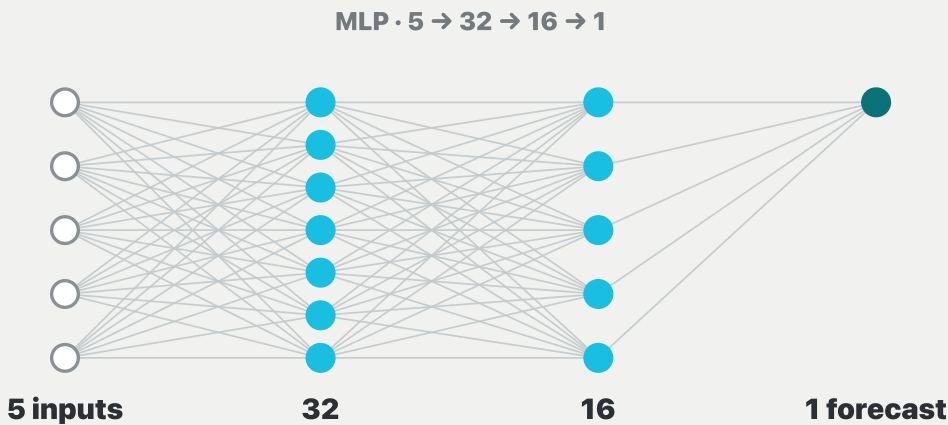


I SPENT 24 HOURS ON A NEURAL NETWORK. ALL I GOT WAS THIS LOUSY T-SHIRT.

My fancy model can learn very complex functions.
"Predict 0% growth" still beat it on mean absolute error (MAE).

EXPLANATION OF THE TARGET VARIABLE
NEXT-QUARTER DOMESTIC DEPOSIT GROWTH

Deposits	\$100m → \$96m
Actual growth	-4%
Zero forecast	0%
Absolute error	4 percentage points



THE CONSTANT PREDICTED 0% GROWTH FOR EVERY BANK.

THE BANK CAN LEND OUT MONEY YOU DEPOSIT. IT DOESN'T KEEP IT ALL IN CASH.

WHEN YOU WITHDRAW, SOME FUNDS MAY STILL BE TIED UP IN LOANS.



CASH IN THE VAULT

\$10



MONEY ALREADY OUT AS LOANS

\$90

A customer withdraws \$15. Cash is \$5 short.

THE BANK'S TREASURY TEAM MUST FIND THE CASH.

HOW MUCH CASH IS ENOUGH?



Required reserve ratio: 0%. That does not mean unlimited lending.

There is no universal “keep 10%, lend 90%” rule. Capital, liquidity and other limits constrain lending.

The full liquidity coverage rule requires eligible liquid assets covering **100% of 30-day stressed net cash outflows.**

CASH

DEPOSITS

LOANS

CAPITAL

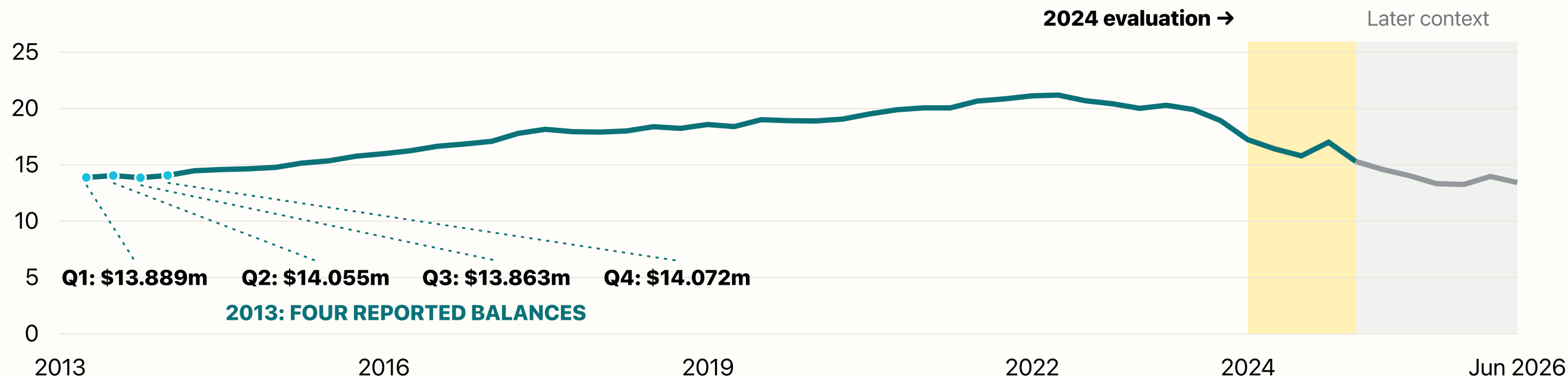
**CUSTOMERS DEPOSIT AND WITHDRAW MONEY.
I'M FORECASTING THE NET CHANGE NEXT QUARTER.**

Example only. Reserve requirement ratios are 0%; separate liquidity rules vary by bank. Liquid assets include eligible securities, not just cash. Source: Federal Reserve, Regulation WW. This project forecasts deposit growth, not bank failure.

"PREDICT 0%" WON. IS THIS THE GRAPH YOU WERE EXPECTING?

2013–2026 · ONE-BANK ILLUSTRATION. Zero won MAE across all test banks, not necessarily this bank.

YOUNG AMERICANS BANK · DOMESTIC DEPOSITS (\$ MILLIONS)

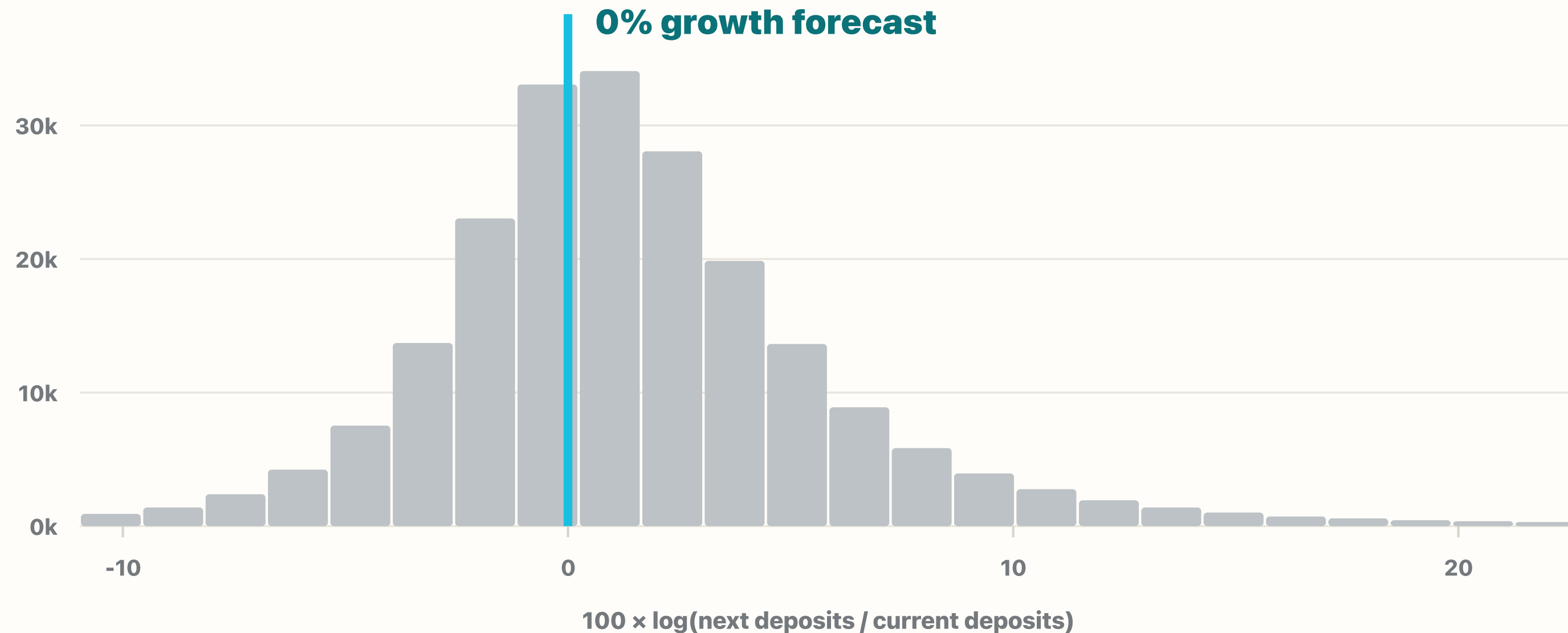


0% growth means "same balance next quarter."
Look at 2013: about \$14 million, quarter after quarter.
"Same again" would be a close guess.

Was little change common across the other banks too? →

Source: FDIC BankFind, Young Americans Bank (CERT 27010), quarterly domestic deposits, 2013–Jun 2026.
2013 is a deliberately stable example. Other years changed more. Dotted lines connect labels to reported balances.

SO WHY WAS 0% GROWTH SO HARD TO BEAT?



A LOT OF THE TRAINING TARGETS LIVE NEAR ZERO.

Central 98% shown · full tails stayed in the experiment · display scale is 100 × log growth

BIG ERRORS GET **SQUARED**. SO THEY CAN PULL THE MODELS AWAY FROM 0%.

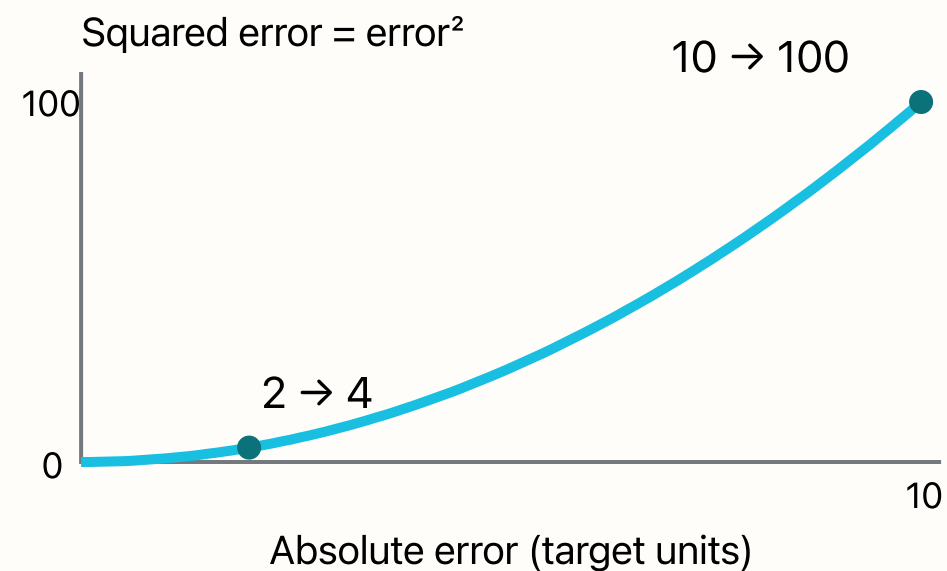
An error 5× as large gets 25× the squared-error penalty.

FIXED BENCHMARK

0% BASELINE

$$\hat{y} = 0$$

Always predicts 0%.
It never adjusts to reduce an error.



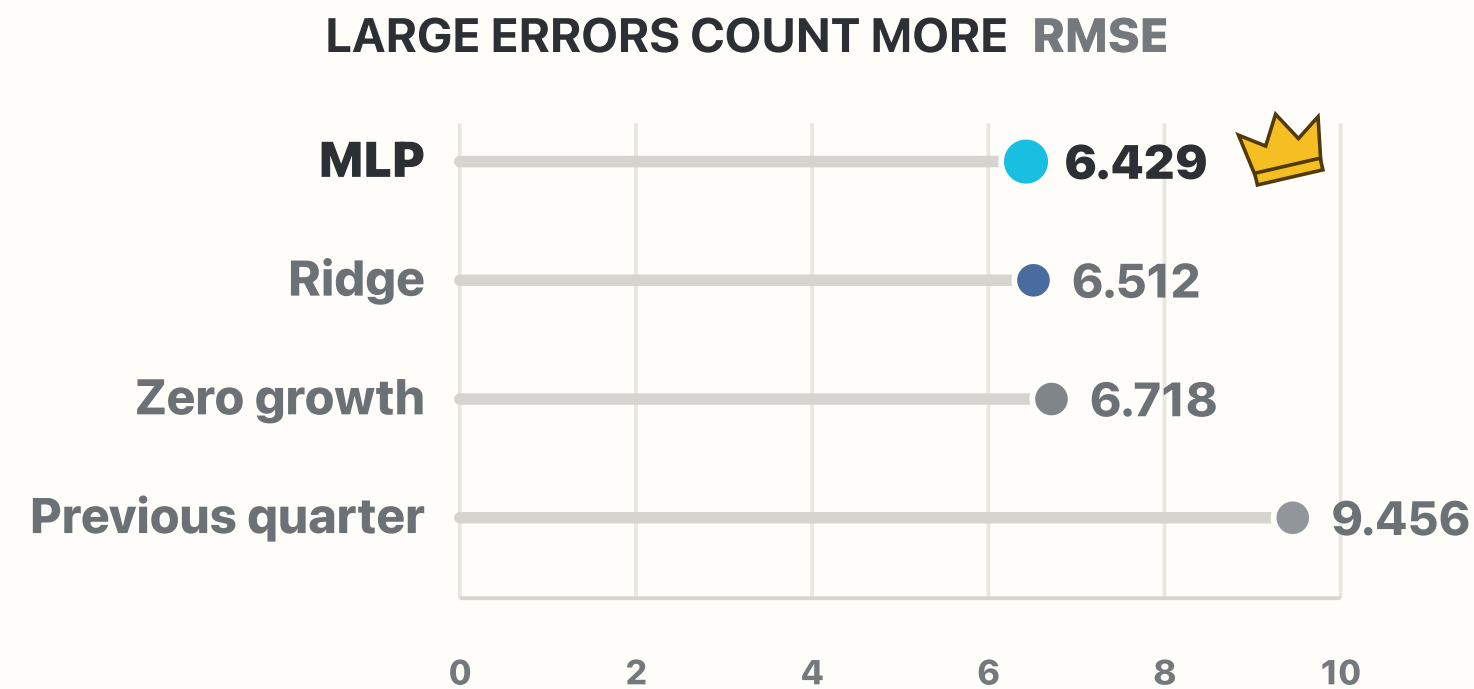
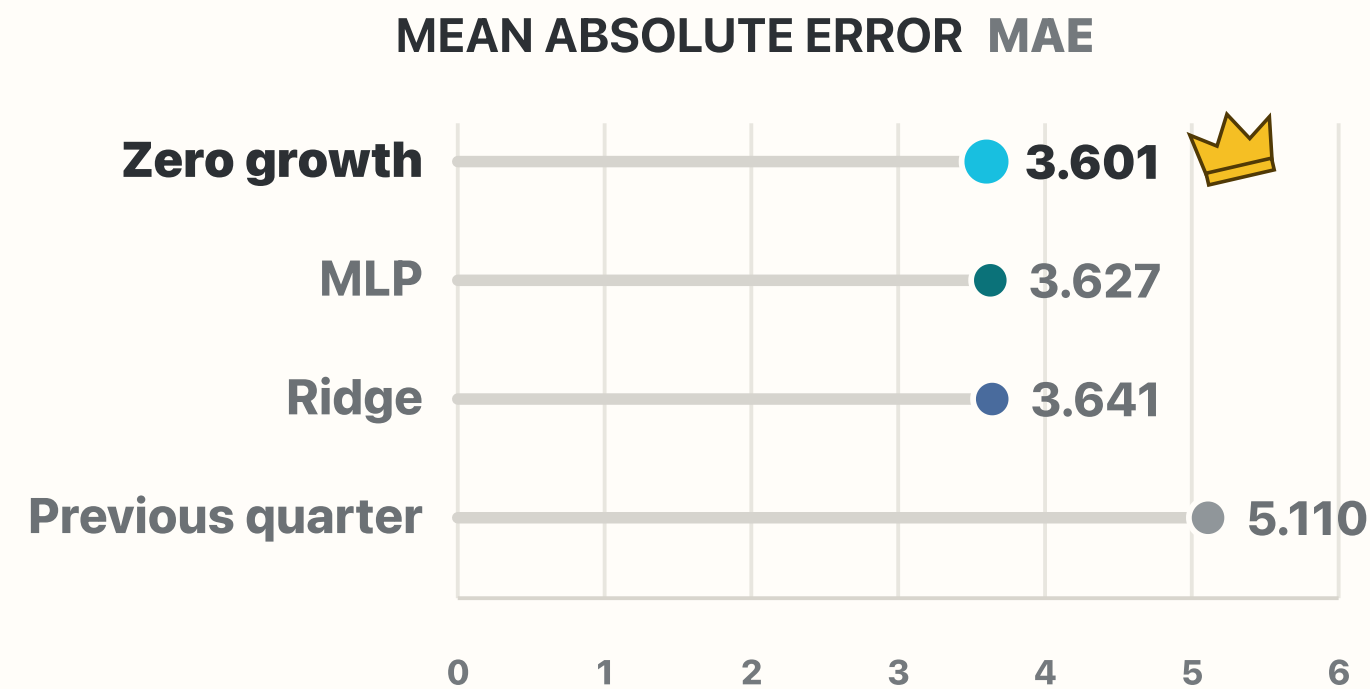
LEARNED MODELS

RIDGE + NEURAL NET

Can move away from 0% to reduce those expensive large training errors.

**A LOWER SQUARED ERROR CAN COME WITH
A SLIGHTLY HIGHER MEAN ABSOLUTE ERROR.**

ZERO WON MAE. THE NEURAL NETWORK WON RMSE.



“WINNER” DEPENDS ON THE ERROR METRIC WE CHOOSE.
MAE averages absolute errors. RMSE squares errors first, so large errors count more.

Same 13,532 evaluation examples · three already-examined 2024 quarters

0% TIES EVERY BANK.

THE LEARNED MODELS TELL ME WHO TO REVIEW FIRST.

★ I am not a financial advisor. I do have a sorted table.

In this March 2024 queue, review Young Americans Bank first: Ridge predicts the largest deposit decline, −4.02%. Check its reports to understand the forecast; this is not a current distress warning.

ORDERED REVIEW QUEUE · SIMPLE LINEAR MODEL (RIDGE) · FIRST 10 OF 454

RANK	BANK	PREDICTED GROWTH	0% BASELINE
01	YOUNG AMERICANS BANK	-4.02%	0% · tied
02	PIONEER COMMERCIAL BANK	-3.32%	0% · tied
03	SUMITOMO MITSUI TR BK U SA	-2.83%	0% · tied
04	OLD GLORY BANK	-2.75%	0% · tied
05	TURTLE MOUNTAIN STATE BANK	-2.11%	0% · tied
06	COLUMBUS STATE BANK	-2.05%	0% · tied
07	FDS BANK	-2.03%	0% · tied
08	BOELUS STATE BANK	-1.96%	0% · tied
09	LEMONT NATIONAL BANK	-1.80%	0% · tied
10	CITIZENS NB OF CROSBYTON	-1.75%	0% · tied

GOAL: FIND MORE LOWEST-GROWTH BANKS WITH THE SAME 454 REVIEWS.

HIGHER IS BETTER.

Random selection

45.4 expected

Linear model (Ridge)

112 found

Neural network

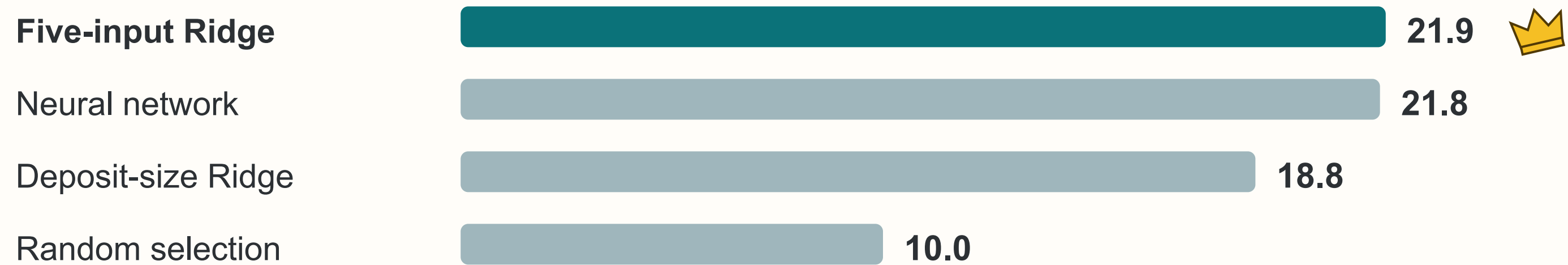
109 found

4,536 eligible banks · March → June 2024. “Lowest-growth” = later in that quarter’s bottom 10% of deposit growth. 112 matches across the full linear-model queue; 109 for the neural net. Review priority is not a distress diagnosis.

OKAY. WE CAN RANK THE BANKS.
CAN DEPOSIT SIZE ALONE MATCH FIVE INPUTS?

No. Size alone found fewer low-growth banks than five-input Ridge.

BANKS FOUND IN THE LOWEST-GROWTH 10% · HIGHER IS BETTER



Low-growth banks found per 100 banks reviewed

ZERO GIVES EVERY BANK THE SAME ANSWER.
IT CANNOT TELL ME WHOM TO REVIEW FIRST.

WHY NOT TRY ANOMALY DETECTION?

HOW DOES PCA CHOOSE BANKS TO REVIEW?

We can review only 10% of banks.

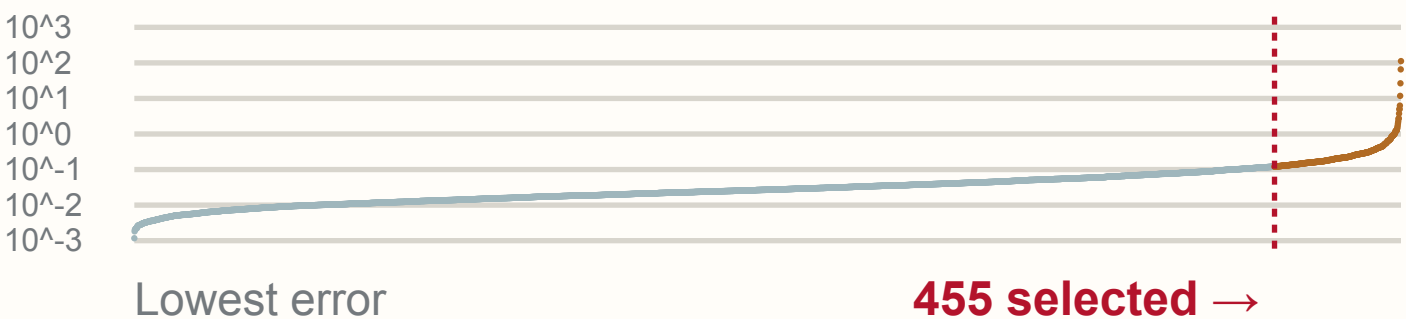
EXAMPLE · Let's try PCA.

It learns to compress past bank histories and rebuild the original numbers.

We review the banks with the biggest errors.

1. RANK ALL 4,548 BANKS

March 2024 · PCA reconstruction error (log scale)



455 banks make the cut.

Does that help us find low-growth banks?

Source: saved March 2024 PCA scores. Unusual histories are not proof of distress; next-quarter growth is evaluated on the next page.

SCOPE

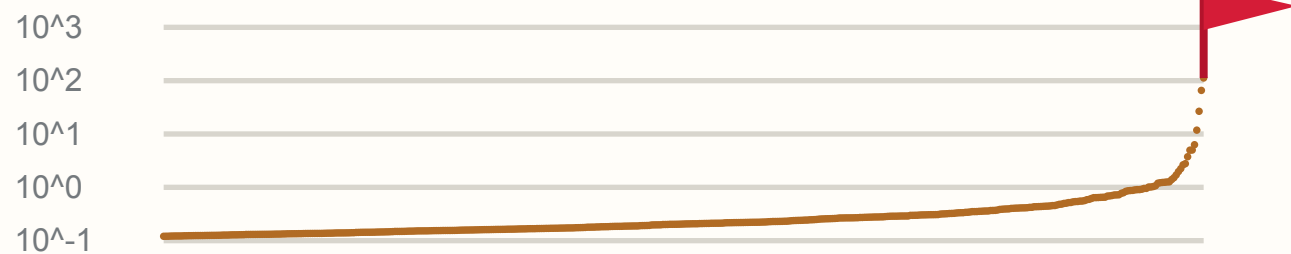
Eight quarters · five financial inputs

ANOMALY MODELS TESTED

PCA, Isolation Forest, and a small autoencoder

2. REVIEW THE TOP 10%

455 banks · reconstruction error (log scale)



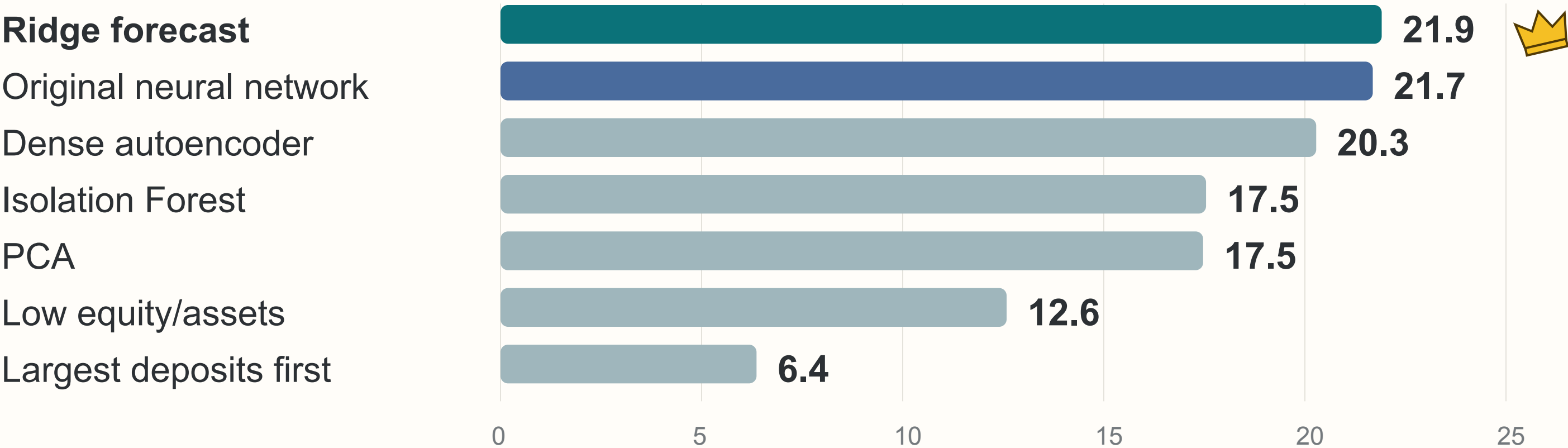
SILVERGATE: THE HIGHEST ERROR

PCA reconstructed its eight-quarter history least accurately. That is why it ranks first.

RIDGE RANKED THE REVIEW LIST BEST HERE.

Low-growth banks found per 100 banks reviewed

Target: next-quarter bottom 10% in deposit growth. Higher is better.



PREDICT ZERO CANNOT RANK BANKS. EVERY FORECAST IS TIED.

1,351 bank-quarter reviews per method · 13,490 eligible rows · three reused 2024 quarters
Primary seed 42 shown. These historical results do not establish future performance.

BUT WAIT... IN THESE VOLATILE ECONOMIC TIMES, WHAT IF WE TRAIN ON ONLY THE RECENT YEARS?

Maybe older examples are less useful now. So I retrained starting in 2020, then 2021.

Model settings held fixed; validated in 2023; scored on 13,532 bank-quarters in 2024.

BACK TO FORECAST ERROR: NEURAL NETWORK MAE (pp) · LOWER IS BETTER

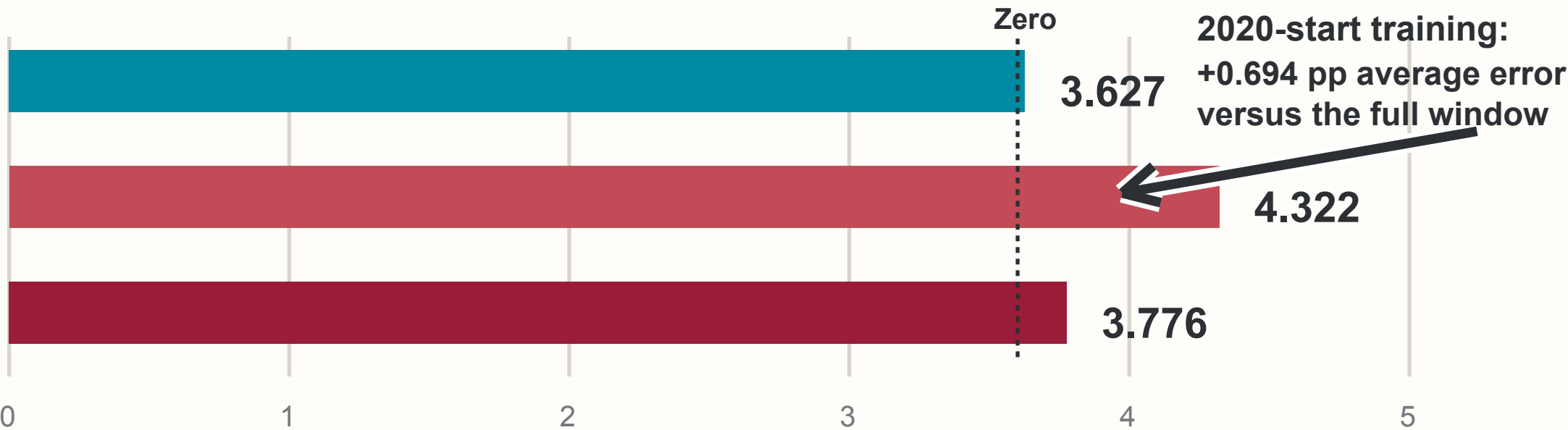
OLDER HISTORY

MORE RECENT HISTORY

2013-Sep 2022

2020-Sep 2022

2021-Sep 2022



ALL THAT TRAINING. ZERO STILL WON.

Zero MAE: 3.601 pp. Lower than every neural-network fit shown.

So what if we keep the history, but let the forecast know the quarter?

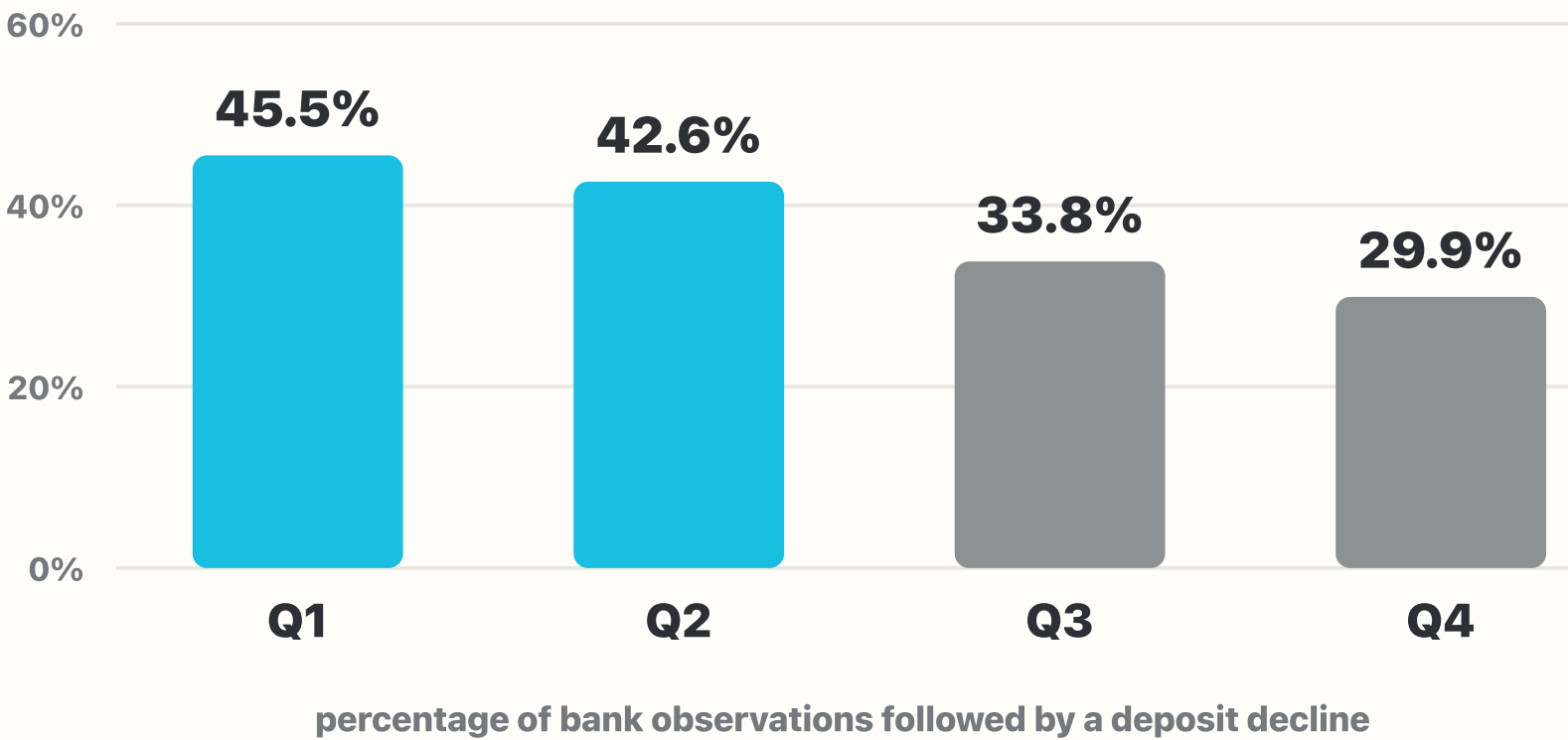


DEPOSIT DECLINES WERE MORE COMMON AFTER Q1 THAN Q4.

Q1: 45.5% of training bank-quarters were followed by a deposit decline. **Q4: 29.9%.**

That is how often declines happened, not “banks lost 45.5% of their deposits.”

HOW OFTEN DID A BANK’S DEPOSITS FALL NEXT QUARTER? · TRAINING YEARS 2013–2022

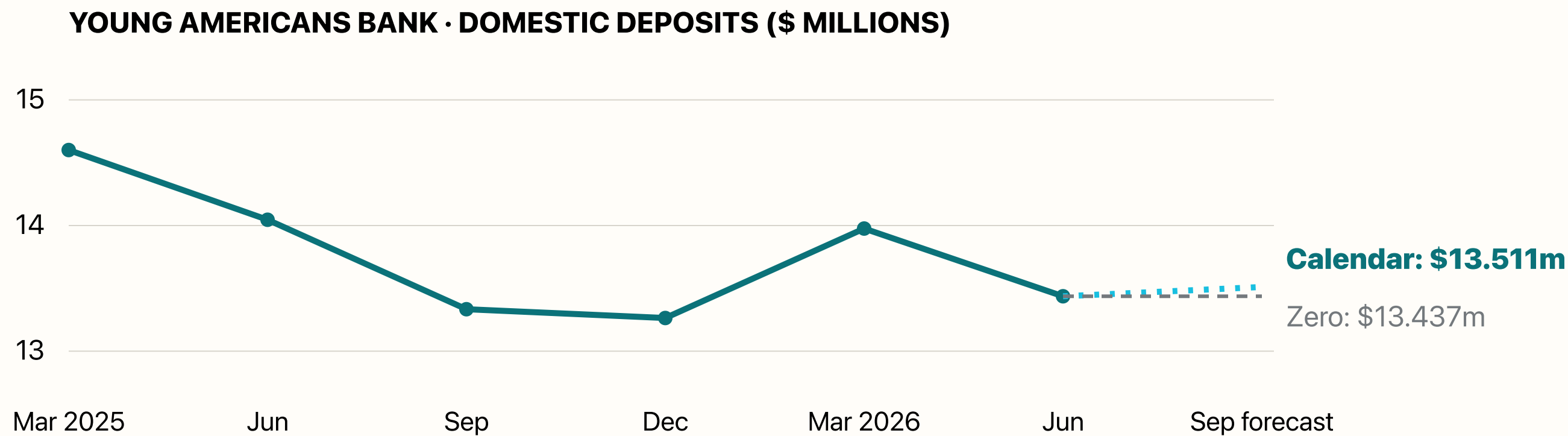


THE STARTING QUARTER MAY HELP SET A GROWTH FORECAST.

For each bar, count bank observations with a next-quarter decline, then divide by all observations starting in that quarter (training data through Sep 2022). Deposit dollars are not added together.

MODEL CHOSEN FOR THE NEXT TEST: THE CALENDAR RULE. NOW WE WAIT FOR Q3 REPORTS.

Q3 ends September 30, 2026. Reports arrive afterward. The saved forecast is +0.550% per bank; zero is the benchmark.



Solid = reported balances. Dotted = calendar forecast (+0.550%). Dashed = no-change forecast.

**WHEN SEPTEMBER REPORTS ARRIVE, COMPARE MEAN ABSOLUTE ERROR.
SAME BANKS. SAME QUARTER. LOWER MAE WINS.**

FDIC BankFind, CERT 27010, retrieved Sep 22, 2026. One-bank illustration of the saved Q2 median rule (+0.550399842%); test MAE across matched banks. Rule developed after seeing 2024. Score matched banks with valid positive June and September domestic deposits; disclose missing reports and merger cases. Example is illustrative. One quarter is not lasting proof.